Exact Tsirelson bound for a three-output Bell functional

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We determine the exact quantum value of the three-output, two-versus-three-input Bell functional that forms the $d = 3$ member of the first family introduced by Perito *et al.* The defining work exhibited a maximally entangled qutrit strategy with value 4 and found a matching upper bound at the $1 + AB$ level of the Navascués–Pironio–Acín hierarchy to numerical precision. Here we give a finite exact certificate over the cyclotomic field $\mathbb{Q}(\omega)$, $\omega^3 = 1$, $\omega \neq 1$. A 35×35 Hermitian Gram matrix has exact rank 16, is positive semidefinite by a rational pivoted-$LDL^\dagger$ factorization, and expands to $4\,\mathbf{1} - I_3$ using only the order-three measurement relations and Alice–Bob commutation. Consequently $I_3 \leq 4$ in arbitrary local dimension, while the qutrit construction proves attainability. The result fixes the Tsirelson value but does not classify equality strategies or by itself imply self-testing, robustness, or device-independent randomness.

I. INTRODUCTION

Bell inequalities turn restrictions on quantum correlations into quantitative operator bounds. For inequalities with several inputs and outputs, sharp quantum values are often first identified through semidefinite relaxations of the quantum moment problem, notably the Navascués–Pironio–Acín (NPA) hierarchy [1–3]. A numerical optimum, however, does not by itself give an exact dimension-independent operator inequality. When a finite NPA dual can be reconstructed as a noncommutative sum of Hermitian squares (SOS), the numerical observation can instead be replaced by an algebraic certificate [4].

Perito *et al.* recently introduced two families of multi-output Bell functionals motivated by optimal global-randomness certification [5]. For their first family, they conjectured an exact quantum value in every output dimension. At $d = 3$, the conjectured value is 4. Their work supplies an explicit value-4 qutrit strategy and a $1 + AB$ NPA upper bound agreeing with 4 to within 10^{-6}, whereas the available analytic estimate is $3\sqrt{2}$ [5]. The exact SOS and self-testing arguments in that work concern its distinct second family.

In this paper we close the value gap for the $d = 3$ member of the first family. The central object is a machine-readable Gram matrix over $\mathbb{Q}(\omega)$ on the complete 35-word Fourier-unitary $1 + AB$ span. Both the coefficient identity and positive semidefiniteness are checked with exact rational arithmetic. The certificate proves the upper bound without a local-dimension cutoff. We keep the stronger equality problem separate: neither the certificate rank nor its kernel alone establishes rigidity or randomness.

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II. BELL SCENARIO AND CONVENTION

Alice chooses $x \in \{0, 1\}$, Bob chooses $y \in \{0, 1, 2\}$, and both outcomes belong to $\mathbb{Z}_3$. For three-output projective measurements $\{M^A_{a|x}\}^2_{a=0}$ and $\{M^B_{b|y}\}^2_{b=0}$, define the Fourier observables

$$A_x = \sum_{a=0}^{2} \omega^a M^A_{a|x}, \qquad B_y = \sum_{b=0}^{2} \omega^b M^B_{b|y}, \tag{1}$$

where $\omega = e^{2\pi i/3}$. They obey

$$A_x^3 = B_y^3 = \mathbf{1}, \qquad A_x^\dagger = A_x^2, \qquad B_y^\dagger = B_y^2. \tag{2}$$

Operators assigned to different parties commute; no relation is imposed between distinct inputs of the same party.

We use the no-dagger convention of Eq. (6) of Ref. [5]:

$$I_3 = \frac{1}{2}\Big[A_0 \otimes (B_0 + B_1 + B_2) + A_1 \otimes (B_0 + \omega B_1 + \omega^2 B_2) + \text{h.c.}\Big]. \tag{3}$$

Appendix A of that reference writes $B_y^\dagger$ instead. The two conventions are equivalent only after recording the Bob outcome inversion

$$M^B_{b|y} \longmapsto M^B_{-b \bmod 3|y}, \qquad B_y \longmapsto B_y^\dagger. \tag{4}$$

All statements below refer to Eq. (3).

III. EXACT CYCLOTOMIC GRAM CERTIFICATE

Let $\mathcal{A}$ be the unital $*$-algebra generated by A_0, A_1, B_0, B_1, B_2, quotiented by Eq. (2) and by $[A_x, B_y] = 0$. Define the ordered word column

$$W = \big(\mathbf{1},\ \{A^k_x\}_{x,k},\ \{B^\ell_y\}_{y,\ell},\ \ \{A^k_x B^\ell_y\}_{x,k,y,\ell}\big)^{\mathsf{T}}, \tag{5}$$

where $x = 0, 1$, $y = 0, 1, 2$, and $k, \ell = 1, 2$, with indices in lexicographic order. Thus W contains $1+4+6+24 = 35$ words.

The supplemental certificate gives a Hermitian matrix $Y \in M_{35}[\mathbb{Q}(\omega)]$. An entry is stored as a rational pair $[a, b]$ representing $a + b\omega$; the overline below denotes entrywise complex conjugation. Its Gram convention is

$$4\,\mathbf{1} - I_3 = \sum_{i,j=1}^{35} \overline{Y_{ij}}\, w_i^\dagger w_j = W^\dagger \overline{Y}\, W. \tag{6}$$

Complex conjugation does not change positive semidefiniteness, so it is enough to certify $Y \succeq 0$.

Two exact checks establish Eq. (6). First, every $w_i^\dagger w_j$ is reduced by moving Alice words past Bob words and combining adjacent powers of the same setting modulo 3. Collecting coefficients gives exactly the constant 4, the six terms of $-I_3/2$ displayed in Eq. (3), and their adjoints; all other reduced-word coefficients vanish. Second, simultaneous exact row and column pivoting produces

$$PYP^\dagger = L\,\mathrm{diag}(d_1, \ldots, d_{16}, 0_{19})L^\dagger, \qquad d_r \in \mathbb{Q}_{>0}, \tag{7}$$

with all d_r stored explicitly as positive rational numbers. The trailing 19×19 Schur complement is identically zero, not merely numerically small. Equation (7) proves $Y \succeq 0$ and $\mathrm{rank}(Y) = 16$. The exact field operations and replay procedure are summarized in the Appendix.

It follows that $\overline{Y} = R^\dagger R$ for a 16×35 matrix R over the algebraic closure, and hence

$$4\,\mathbf{1} - I_3 = \sum_{r=1}^{16} P_r^\dagger P_r, \qquad P_r = \sum_i R_{ri} w_i. \tag{8}$$

This immediately gives the upper bound in every representation of the frozen relations.

TABLE I. Exact checks encoded in the supplemental certificate and replayed by the independent verifier. No tolerance enters these checks.

Quantity	Exact result
Word count	35
Gram-matrix dimension	35×35
Coefficient mismatches in Eq. (6)	0
Positive $LDL^\dagger$ pivots	16
Identically zero remainder dimension	19
Gram rank	16
Trace of Y	4

IV. ATTAINABILITY AND EXACT QUANTUM VALUE

For completeness, consider the normalized maximally entangled qutrit state

$$|\Phi_3\rangle = \frac{1}{\sqrt{3}} \sum_{k=0}^{2} |kk\rangle, \tag{9}$$

and generalized Pauli operators

$$Z = \sum_{k=0}^{2} \omega^k |k\rangle\langle k|, \qquad X = \sum_{k=0}^{2} |k+1\rangle\langle k|. \tag{10}$$

Take $A_0 = Z$, $A_1 = X$, and [5]

$$B_y = \frac{1}{3}\left(2Z^2 + 2\omega^{2y} X - \omega^{y+1} X^2 Z\right). \tag{11}$$

Direct multiplication in $\mathbb{Q}(\omega)$ gives $B_y^\dagger B_y = B_y^3 = \mathbf{1}$, so the inverse Fourier transform of each B_y is a valid three-projector measurement. The same arithmetic gives

$$\sum_{y=0}^{2} B_y = 2Z^2, \qquad \sum_{y=0}^{2} \omega^y B_y = 2X. \tag{12}$$

Because $|\Phi_3\rangle$ is stabilized by $Z \otimes Z^2$ and $X \otimes X$, Eqs. (3) and (12) yield

$$\langle \Phi_3 | I_3 | \Phi_3 \rangle = 4. \tag{13}$$

Combining Eqs. (8) and (13) proves the main result.

Theorem.—For every tensor-product quantum strategy on arbitrary local Hilbert spaces with the three-output projective measurements of Eq. (1),

$$\mathrm{Tr}(\rho I_3) \le 4. \tag{14}$$

The bound is attained by the two-qutrit strategy (9)–(11); therefore the Tsirelson value of Eq. (3) is exactly 4.

The operator identity actually uses only cross-party commutation, so the same upper bound also holds for commuting-operator representations of these relations. Attainability is already finite dimensional.

V. COMPARISON AND LIMITATIONS

The certificate lies on the same $1 + AB$ word span that produced the published numerical upper bound [5], but its logical status is different. The matrix entries, coefficient comparison, and $LDL^\dagger$ pivots are all exact. Floating-point semidefinite optimization was used only to locate a rational point in the affine Gram family and is not used in accepting the identity or positivity. The resulting inequality therefore has no numerical error bar and no local-dimension truncation.

This conclusion is deliberately narrower than global-randomness certification. If a strategy saturates the bound, the sixteen polynomials in Eq. (8) annihilate the state on its support. Turning those kernel conditions into a classification up to local isometries would require a separate rigidity argument. The 19-dimensional null space of the Gram matrix is not such a classification. We consequently make no claim about self-testing, robustness, uniqueness of the behavior, or the guessing probability of the extended functional. Numerical robustness calculations for the surrounding randomness protocol address a complementary question [6].

Nor does the present certificate establish the conjectured first-family value for $d > 3$. Its field, word list, and symmetry reduction are specialized to $\omega^3 = 1$. Generalizing it requires a new dimension-dependent construction rather than a relabeling of Eq. (6).

VI. CONCLUSION

We have converted the numerical value-4 observation for a three-output Bell functional into a finite exact noncommutative SOS. A rank-16 cyclotomic Gram matrix proves $I_3 \leq 4$ in arbitrary dimension, and the known qutrit strategy attains the bound. This supplies a rigorous baseline for future analyses of equality and randomness while leaving those stronger questions open.

Appendix A: Exact replay of the certificate

This appendix records the algebra used by the supplied verifier. Elements of $\mathbb{Q}(\omega)$ are represented by pairs $(a, b) \in \mathbb{Q}^2$ with $a + b\omega$. From $\omega^2 = -1 - \omega$,

$$(a, b)(c, d) = (ac - bd,\ ad + bc - bd), \tag{A1}$$

$$\overline{(a, b)} = (a - b, -b). \tag{A2}$$

Thus every operation required by the proof reduces to integer and rational arithmetic.

Each word is stored as an Alice sequence and a Bob sequence. The reducer combines adjacent occurrences of one setting modulo 3 and deletes exponent 0. Alice and Bob sequences are maintained separately, implementing only their mutual commutation. For each canonical reduced word u, the verifier forms

$$c(u) = \sum_{i,j:\, \mathrm{red}(w_i^\dagger w_j) = u} \overline{Y_{ij}}. \tag{A3}$$

The target map has

$$c(\mathbf{1}) = 4, \tag{A4}$$

$$c(A_0 B_y) = c(A_0^2 B_y^2) = -\tfrac{1}{2}, \tag{A5}$$

$$c(A_1 B_y) = -\tfrac{1}{2}\omega^y, \tag{A6}$$

$$c(A_1^2 B_y^2) = -\tfrac{1}{2}\omega^{-y}, \tag{A7}$$

for $y = 0, 1, 2$, and zero for every other word. Exact dictionary equality therefore verifies the full operator identity, rather than selected moments or a trace.

For positivity, the verifier performs simultaneous row and column swaps, takes a strictly positive rational diagonal pivot, and updates the remaining Schur complement in $\mathbb{Q}(\omega)$. After sixteen pivots, every entry in the remaining 19×19 block is exactly $(0, 0)$. This is the factorization in Eq. (7). The machine-readable matrix, permutation, positive pivots, and independent replay script are supplied with the manuscript.

DATA AND CODE AVAILABILITY

No experimental or external numerical data were used. The machine-readable exact Gram matrix and its rational $LDL^\dagger$ data are provided in `exact-sos-certificate.json`; `verify_exact_sos_independent.py` reconstructs all 35 words, checks every coefficient, and repeats the exact positivity test. The included reproducibility instructions specify the environment and command. NumPy is used only for a non-certifying floating-point eigenvalue diagnostic.

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OpenAI Codex (gpt-5.6-sol) was used substantively for literature synthesis, scientific reasoning, exact-algebra code development and debugging, and manuscript drafting. It was directed to preserve the stated Bell-operator convention and to accept the central result only after exact arithmetic checks; the supplied independent implementation rechecks the complete coefficient identity and positive-semidefinite factorization. The AI system is not an author, and responsibility for final verification and any submission rests with the human author.

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